BSAN 400 Introduction to Machine Learning

Lecture 2. Machine Learning Basics[1]

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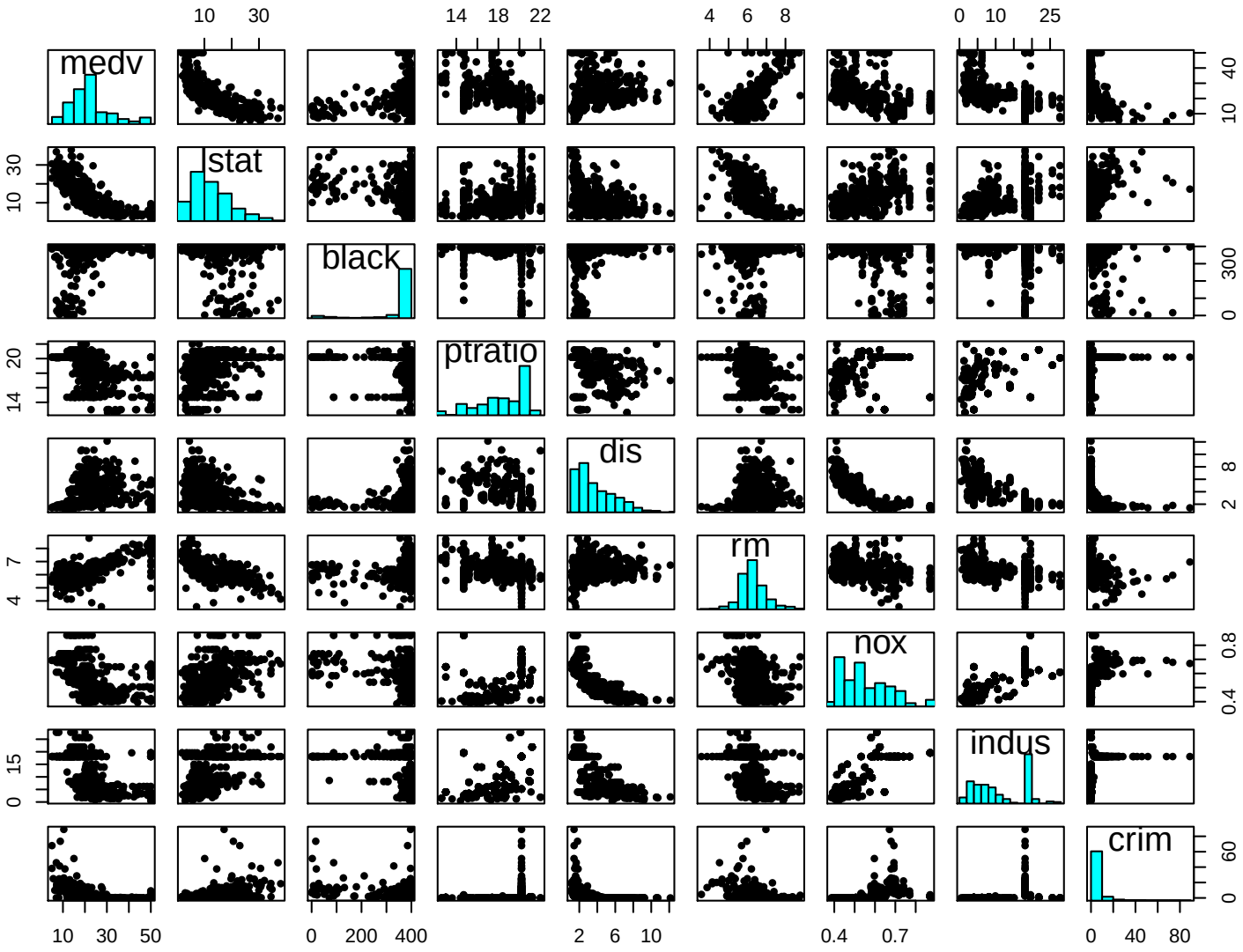
[1] Partially based on Hastie, et al. (2009) ESL, and James, et al. (2013) ISLR

Machine Learning in Different Fields

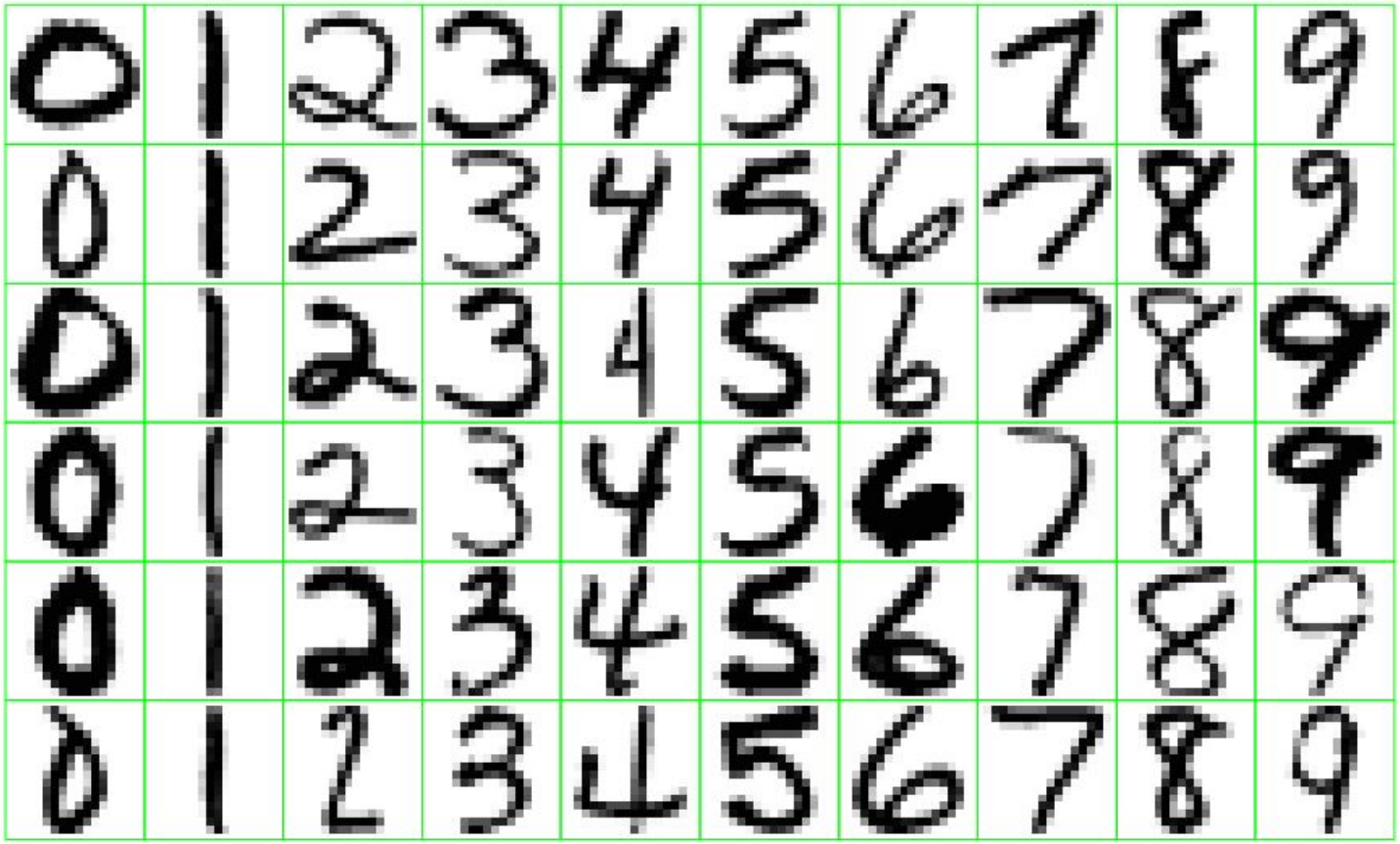
Learning from data is essential in different scientific disciplines

- Predict stock returns in next six months based on historical data;
- Predict the probability of a loan default based on customer's information and historical records;
- Identify certain diseases based on medical image;
- Identify handwritten digits from image;
- Facial recognition;
- Natural language processing;
- Cluster customers based on their purchase behavior and other information

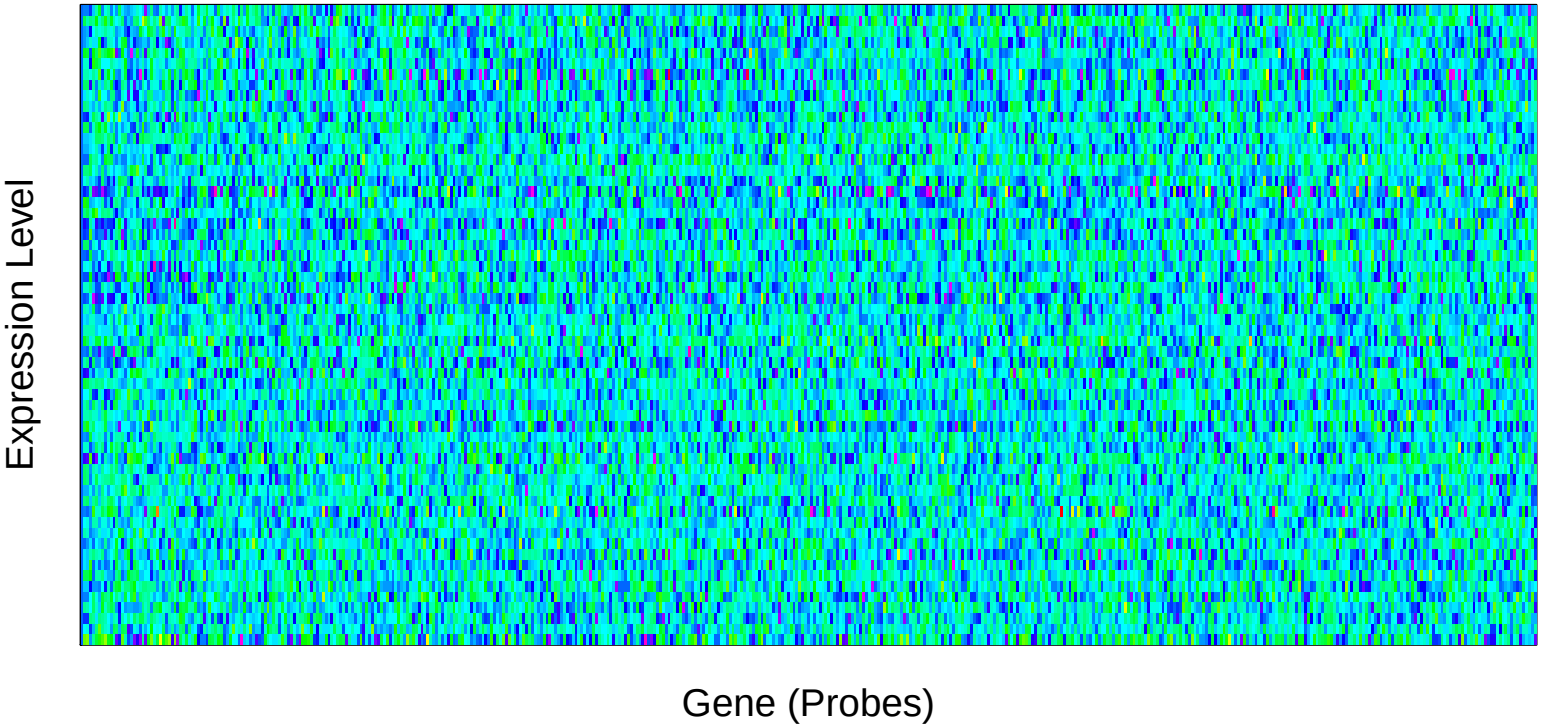
Examples



Examples



Examples



Learning Types

- **Supervised learning**
  - There is specific response you need to predict
- **Supervised learning**
  - No response, instead, you need to create response based on some patterns
- **Semi-supervised learning**
  - Mixture of both

Supervised Learning

- Suppose we observe data Y_i and $\mathbf{x}_i = (X_{i1}, \ldots, X_{ip})$ for $i = 1, \ldots, n$.
- Y is the outcome (or response, dependent variable, target), and $\mathbf{x}$ is predictor (or independent variables, covariates, features, inputs)
- The learning problem can be modeled as

$$Y_i = f(\mathbf{x}_i) + \epsilon_i$$

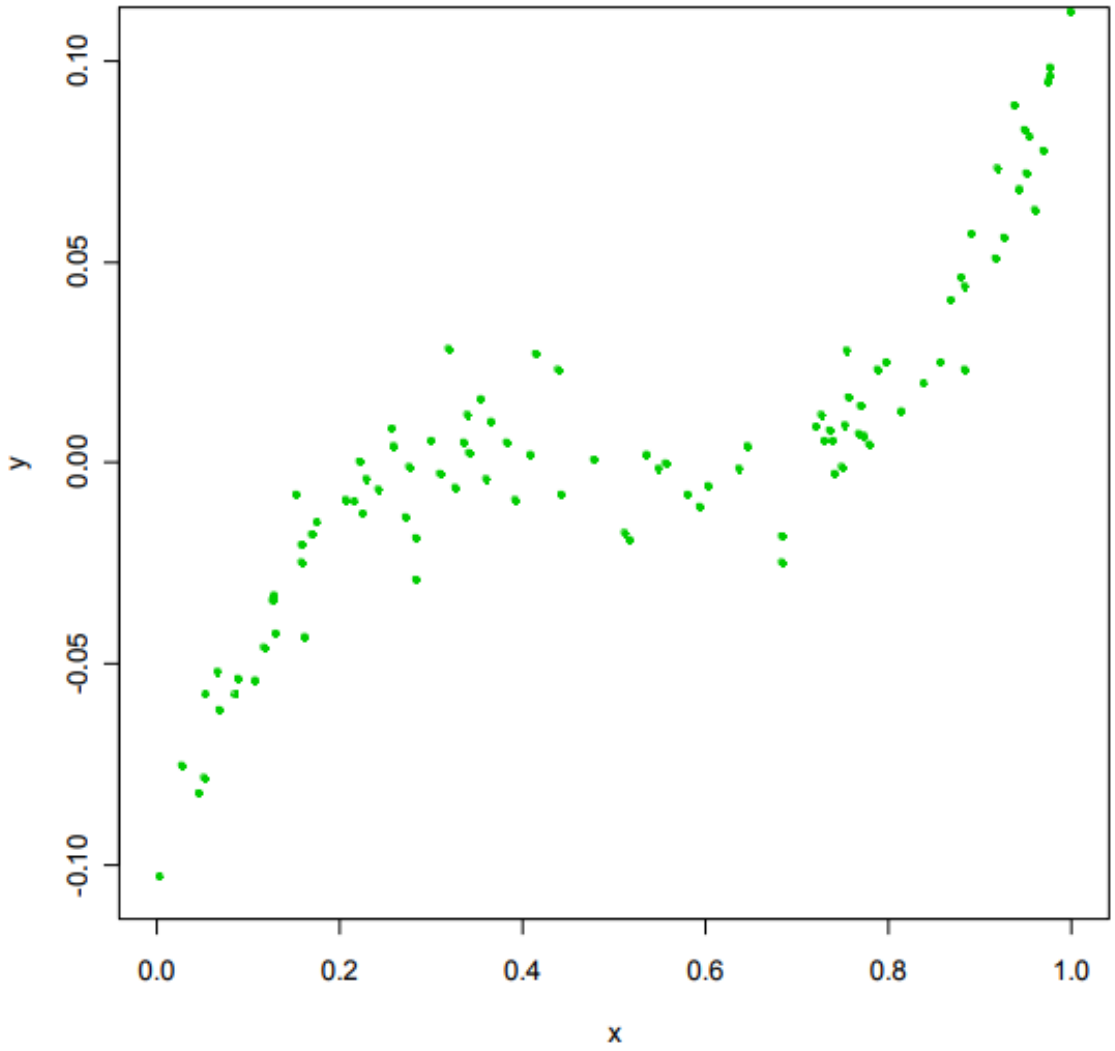
where $f(\cdot)$ is unknown function, and ϵ is random error.

- $\epsilon = Y - f(x)$ is *irreducible* error, or systematic error
- The ultimate goal is to estimate the function $f(\cdot)$

$$\hat{f}(\mathbf{x}) = \mathbb{E}(Y|\mathbf{x}) = \hat{Y}(\mathbf{x})$$

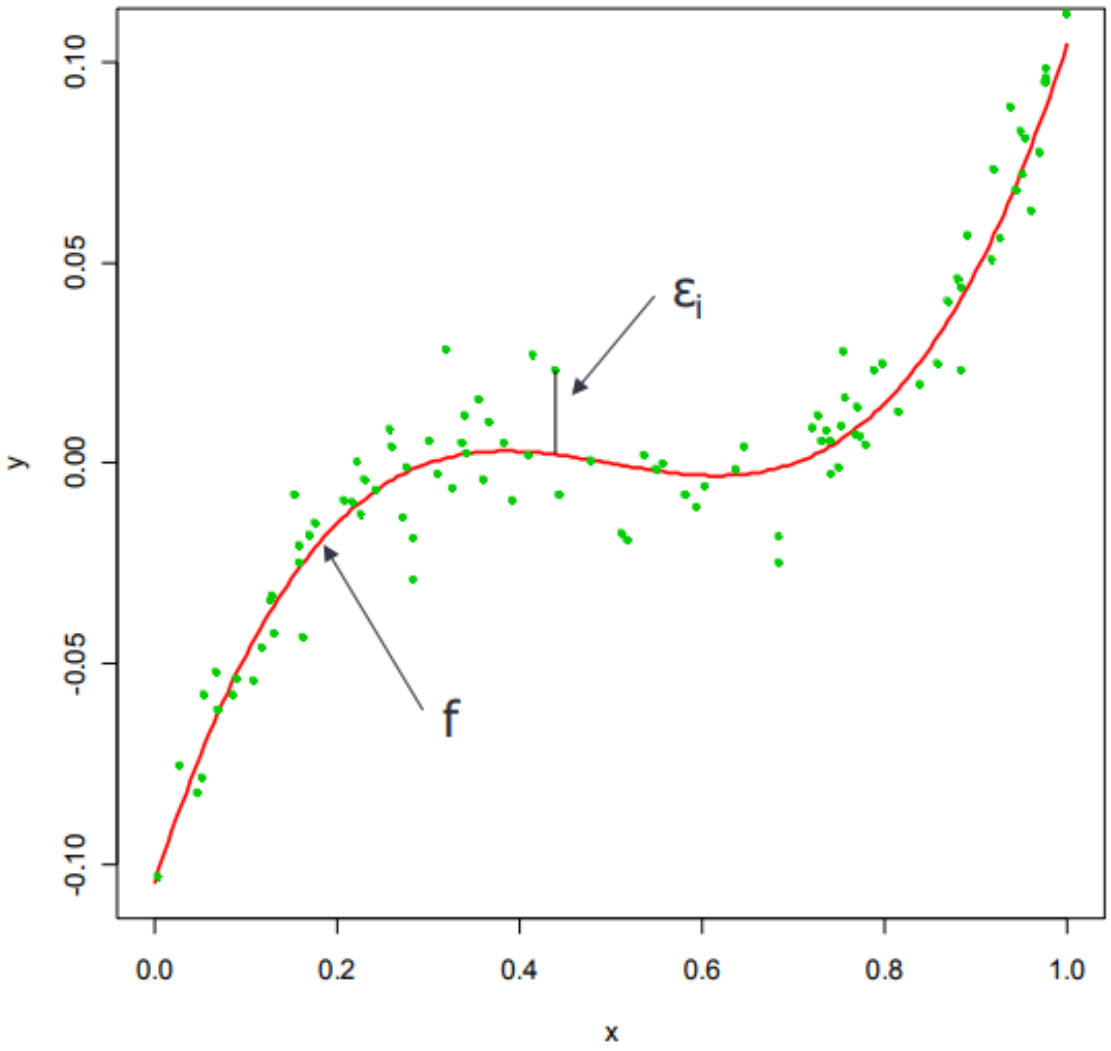
- **Prediction**: given $\mathbf{x}$, we can make valid prediction of Y. That is, to make the **prediction error** as small as possible.
- **Inference**: Which $x's$ are important, and how the change of $\mathbf{x}$ affect Y?

Simple Example[2]



[2] Prof. Abbass Al Sharif's lecture notes

Simple Example[2]



[2] Prof. Abbass Al Sharif's lecture notes

Regression and Classification

Regression:

- Response variable is continuous
- e.g., stock return, housing price, temperature

Classification:

- Response variable is categorical
- e.g., {A, B, C}, {dog, cat}, {0, 1}

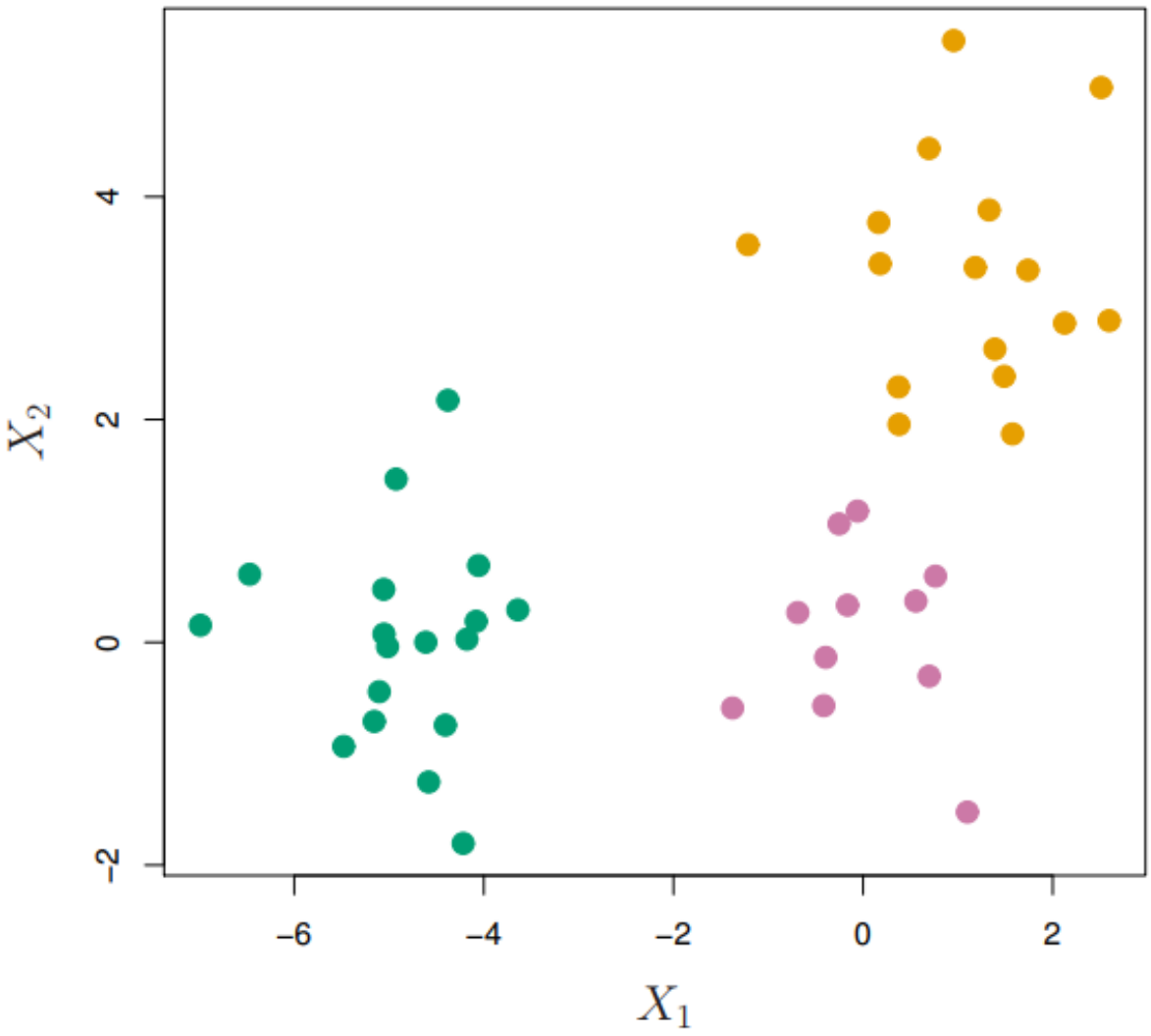
Learning methods:

- Parametric methods, e.g., Maximum likelihood
- Nonparametric methods, e.g., decision tree, kernel method

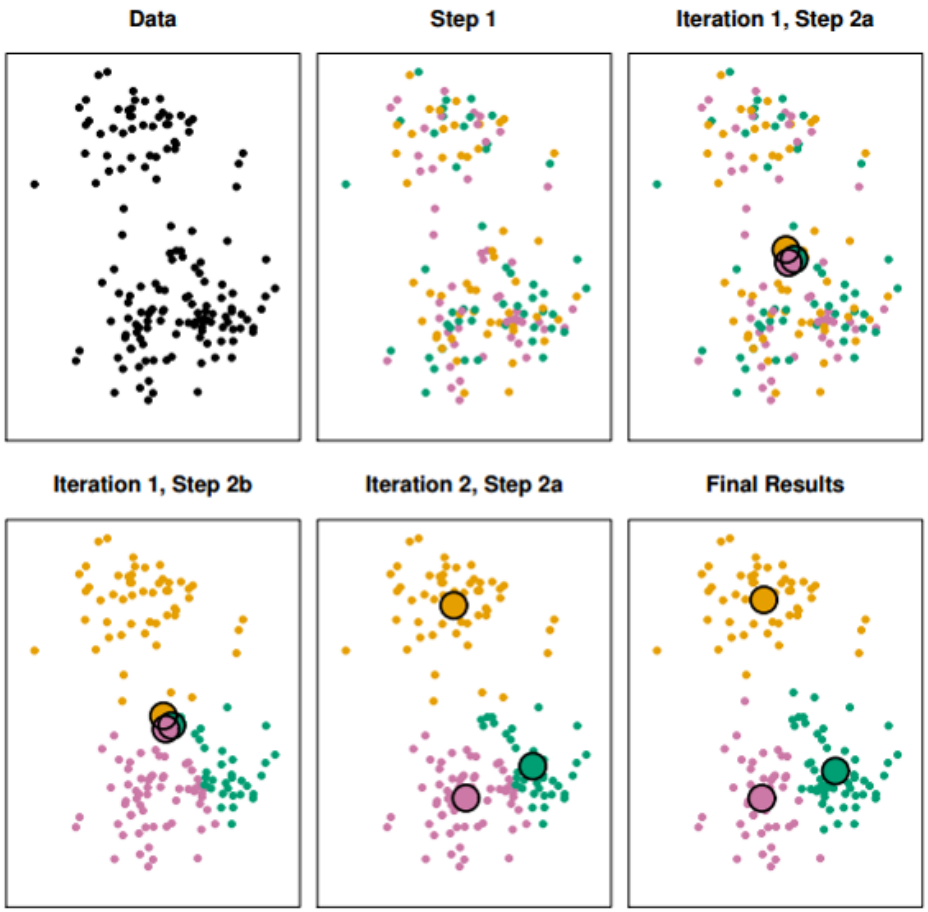
Unsupervised Learning

- Data is **unlabeled** (no Y's)
- Uncover patterns, groups among X's
- Subjective, no simple goal such as prediction
- Examples: Recommendation systems, clustering, principle component analysis (PCA), association rules

K-means Clustering



How it works



- Here is a very good animation to illustrate k-means clustering algorithm. [link]

Supervised Learning – Least Squares

- Linear regression model

$$Y = \beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p + \epsilon$$

- The estimated model is (using matrix format)

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$$

- Solve $\hat{\beta}$ using least square

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} (y_i - \boldsymbol{\beta}^T \mathbf{x}_i)^2$$

- Exercise: derive the analytical solution using matrix format.

Supervised Learning – Nearest Neighbor

- No model assumption
- Nonparametric method

$$\hat{Y}(\mathbf{x}) = \frac{1}{k} \sum_{\mathbf{x}_i \in N_k(\mathbf{x})} y_i$$

  where $N_k(\mathbf{x})$ is the neighborhood of $\mathbf{x}$ defined by the k closest points $\mathbf{x}_i$.
- k determines the flexibility of the model (should larger k or smaller k results in more flexible model?)
- How to define the neighbor? (How to find closest points to $\mathbf{x}_i$?)
  - Similarity measures

An example[3]

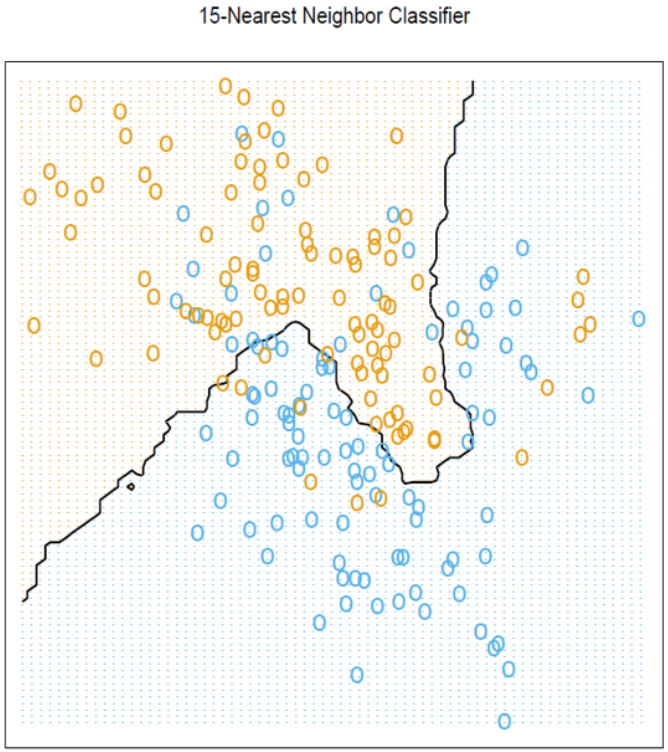


FIGURE 2.2. *The same classification example in two dimensions as in Figure 2.1. The classes are coded as a binary variable* (BLUE = 0, ORANGE = 1) *and then fit by 15-nearest-neighbor averaging as in (2.8). The predicted class is hence chosen by majority vote amongst the 15-nearest neighbors.*

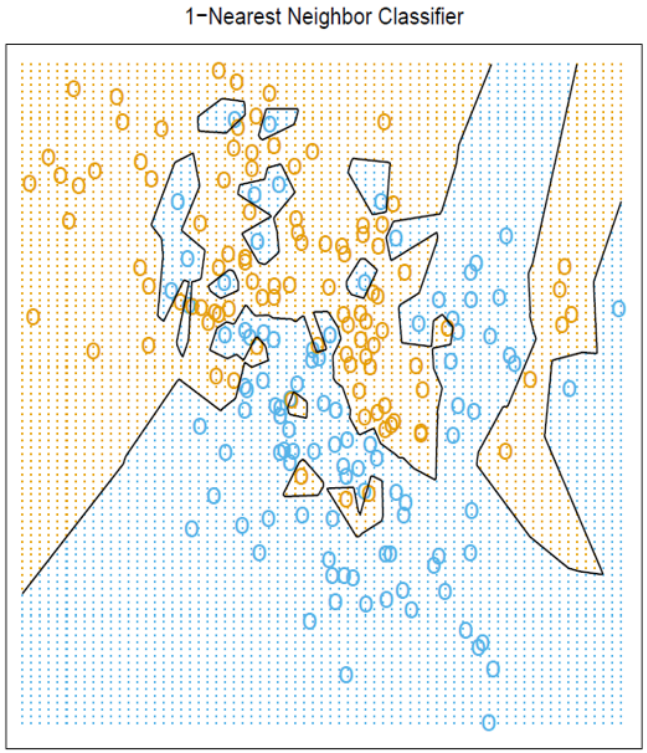
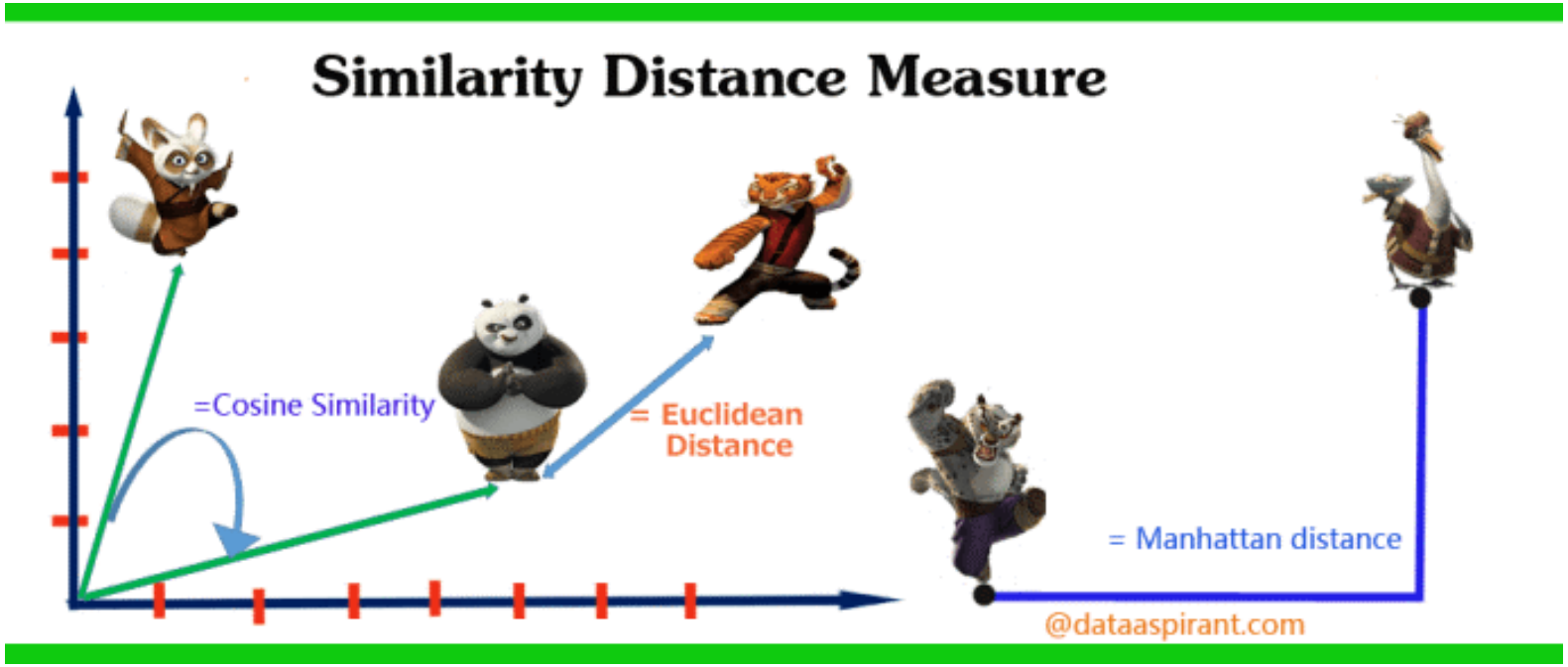


FIGURE 2.3. *The same classification example in two dimensions as in Figure 2.1. The classes are coded as a binary variable* (BLUE = 0, ORANGE = 1), *and then predicted by 1-nearest-neighbor classification.*

[3] ESL pg.15-16

Distance-based Similarity Measures



Model Assessment

How do we know if the estimated model $\hat{f}(x)$ is **useful**?

- We never know the true $f(x)$
- Split sample to training and testing sets
- Train the model (learning algorithm) based on training sample by minimizing *training error*
- Apply the estimated model on the testing sample to calculate the *prediction (testing) error*
- We care more about testing error rather than training error

Prediction Error

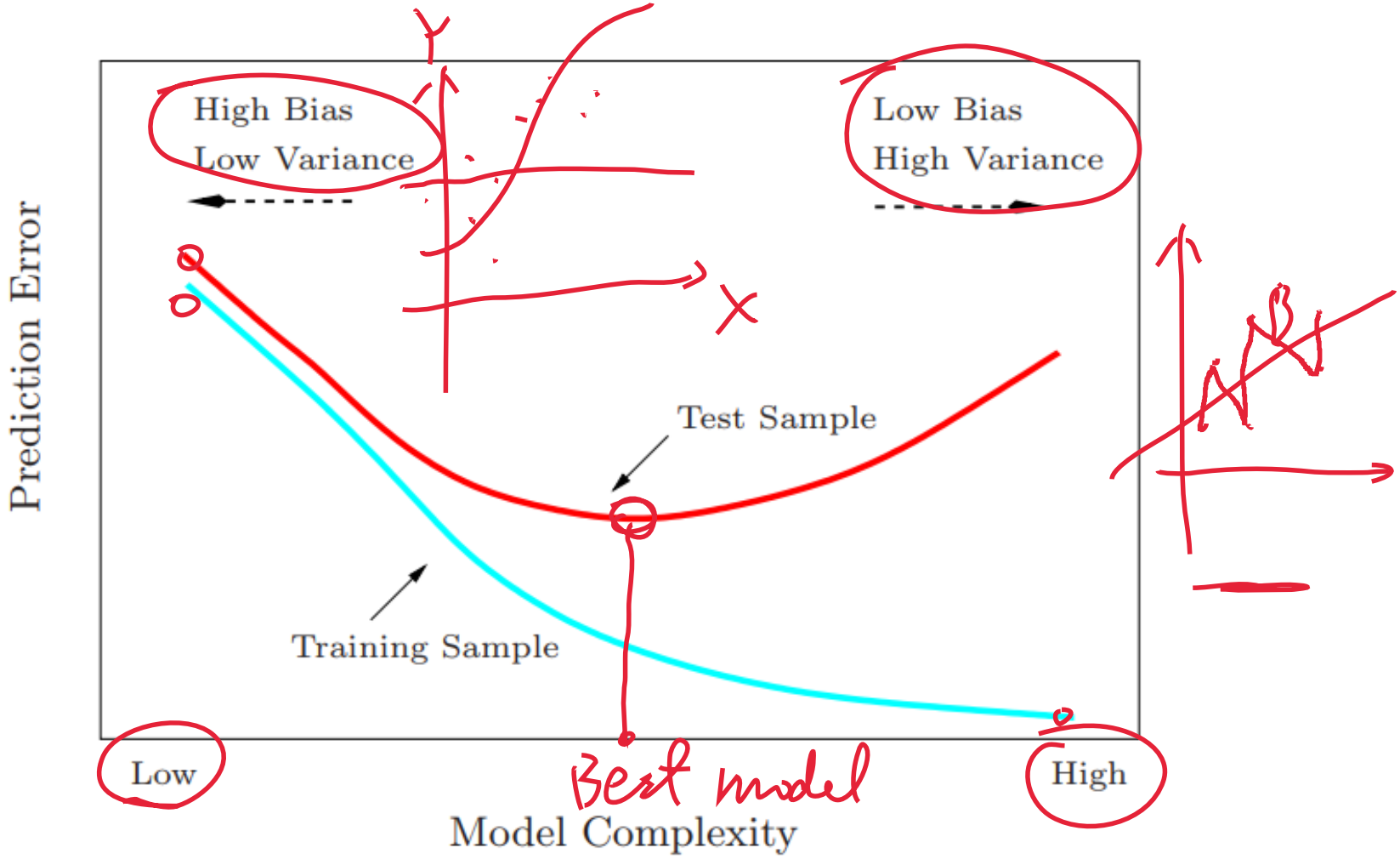
One of the most common metrics is the *mean squared error*

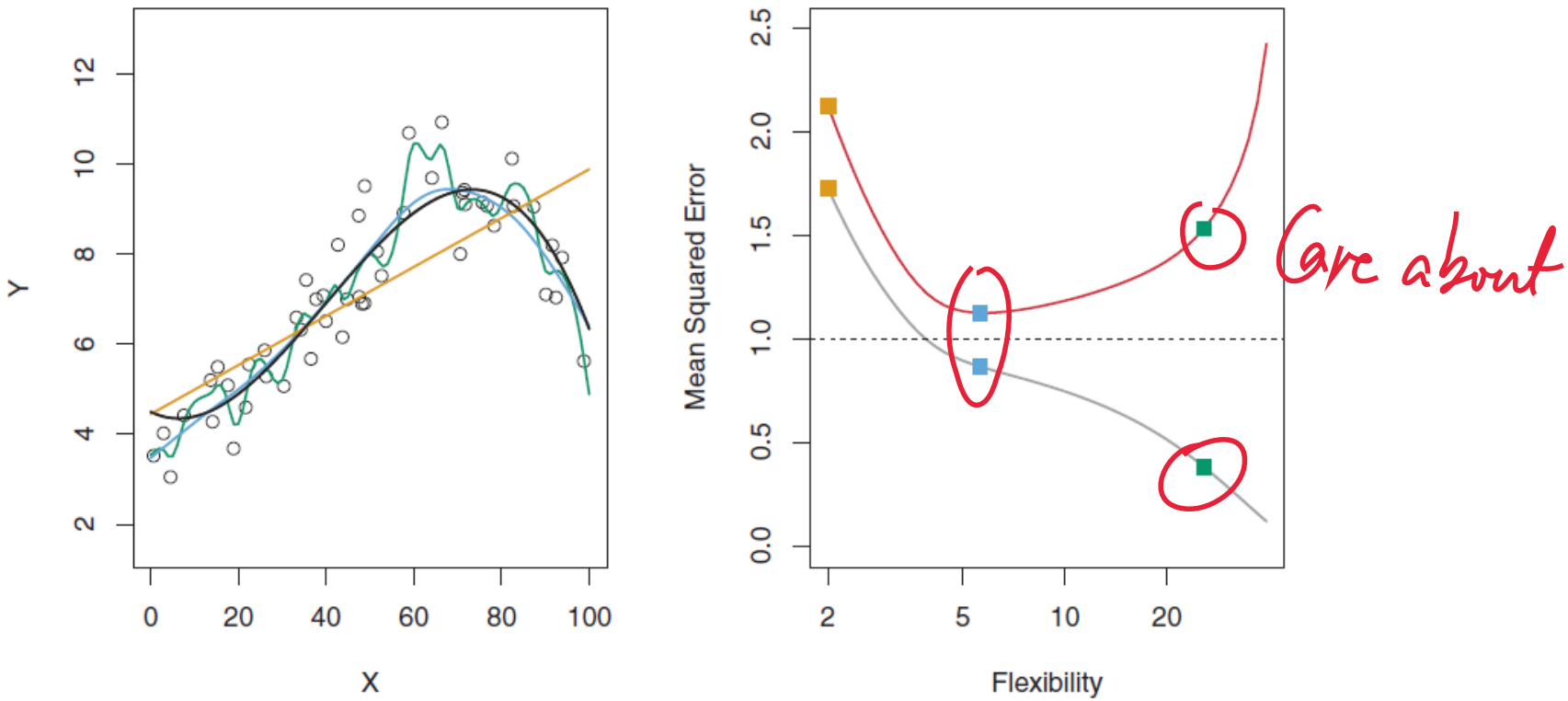
- Denote training set $Tr = \{x_i, y_i\}_1^N$, and testing set $Te = \{x_i, y_i\}_1^M$

$$\text{MSE}_{Tr} = \frac{1}{N} \sum_{i \in Tr} (y_i - \hat{f}(x_i))^2$$

$$\text{MSE}_{Te} = \frac{1}{M} \sum_{i \in Te} (y_i - \hat{f}(x_i))^2$$

- Training error, MSE_{Tr}, may be biased due to overfitting
- In this course, we denote *MSE* as training error, and *MSPE* as testing error


High Bias
Low Variance
Low Bias
High Variance
Prediction Error
Test Sample
Training Sample
Low
High
Model Complexity
Best model



Left: Orange, blue and green curves correspond to fits of different flexibility;
Right: Red curve is MSPE (testing error), grey curve is MSE (training error).

K-Fold Cross-validation

- Instead of doing training vs. testing once, we do it K times

	Folder 1	Folder 2	Folder 3	Folder 4	Folder 5
Pred. Err 1	Test	Train	Train	Train	Train
Pred Err 2	Train	Test	Train	Train	Train
⋮	Train	Train	Test	Train	Train
	Train	Train	Train	Test	Train
Pred Err 5	Train	Train	Train	Train	Test

- Use 2,3,4,5 as training and 1 as testing
- Use 1,3,4,5 as training and 2 as testing
- Keep doing this loop...
- Average 5 testing errors, that is CV score $= \frac{1}{5}(PE1 + \cdots + PE5)$

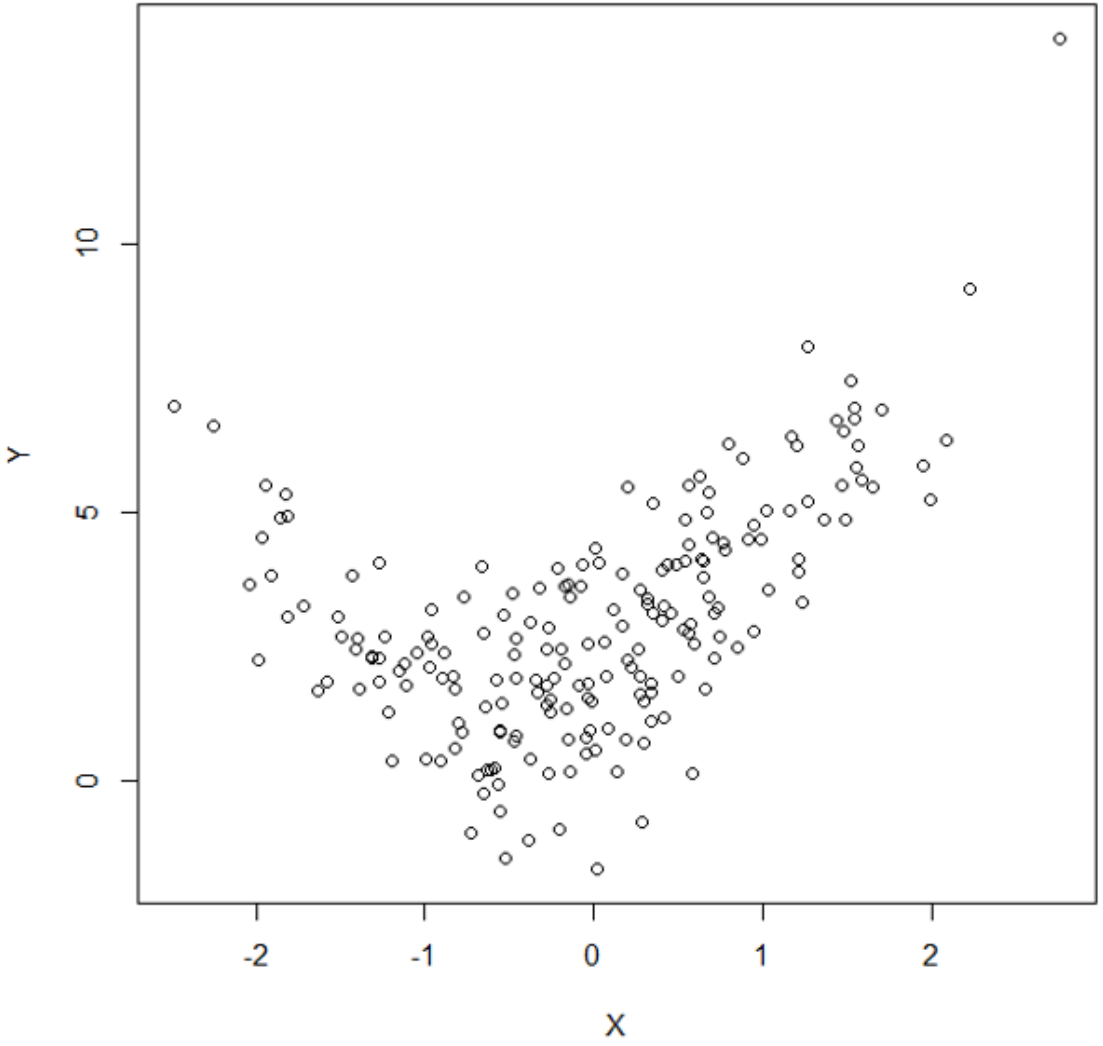
Leave-one-out Cross-validation

- By the name, it requires to repeat training-testing procedure n times
- However, for least square linear model, there is a short cut that makes LOOCV the same that of a single model fit

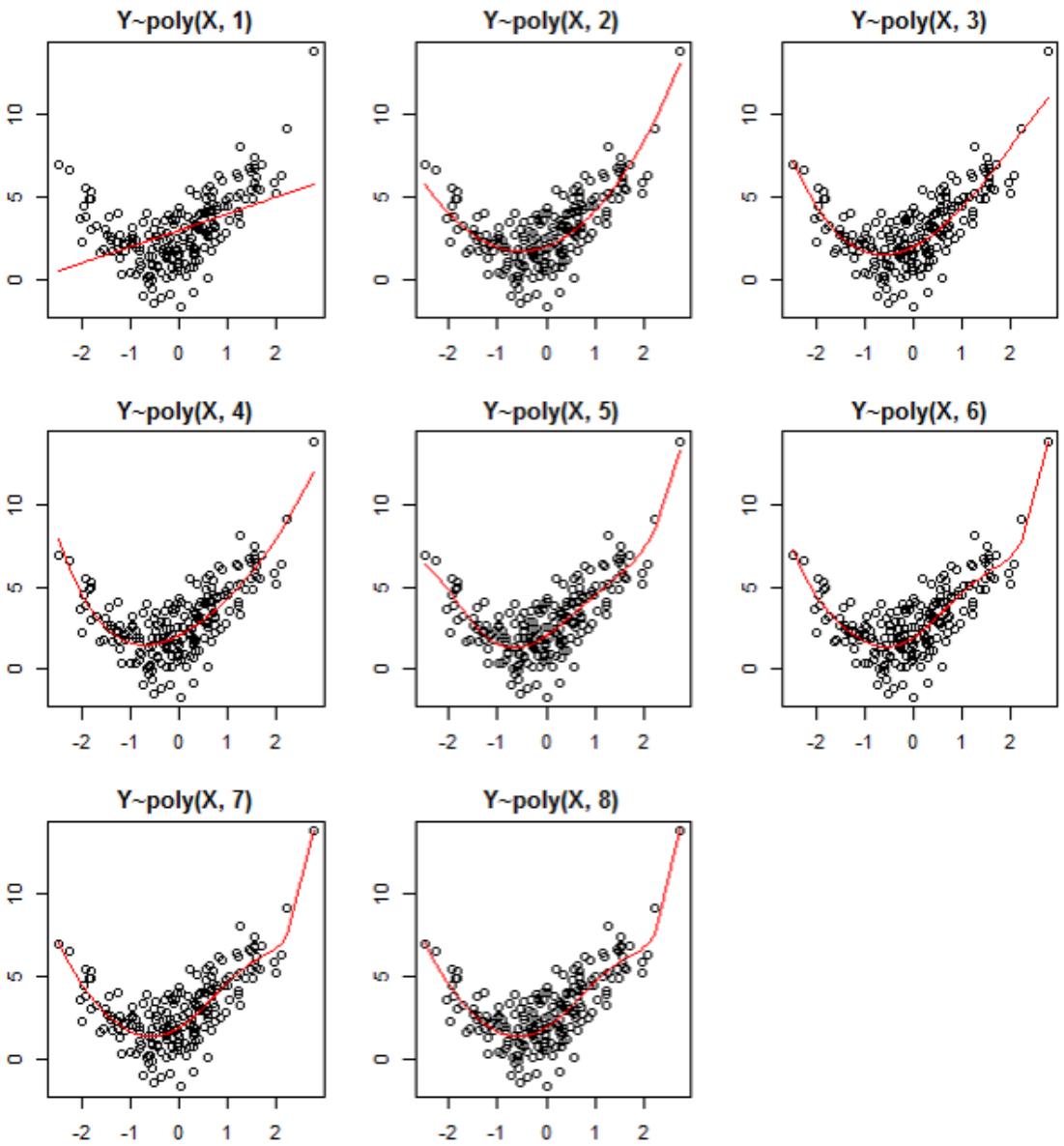
$$CV_n = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \hat{y}_i}{1 - h_i}\right)^2$$

  where h_i is the diagonal element of "hat" matrix.
- In general, the estimates from LOOCV are highly correlated hence their average can have high variance
- In practice, $K = 5$ or 10 is recommended

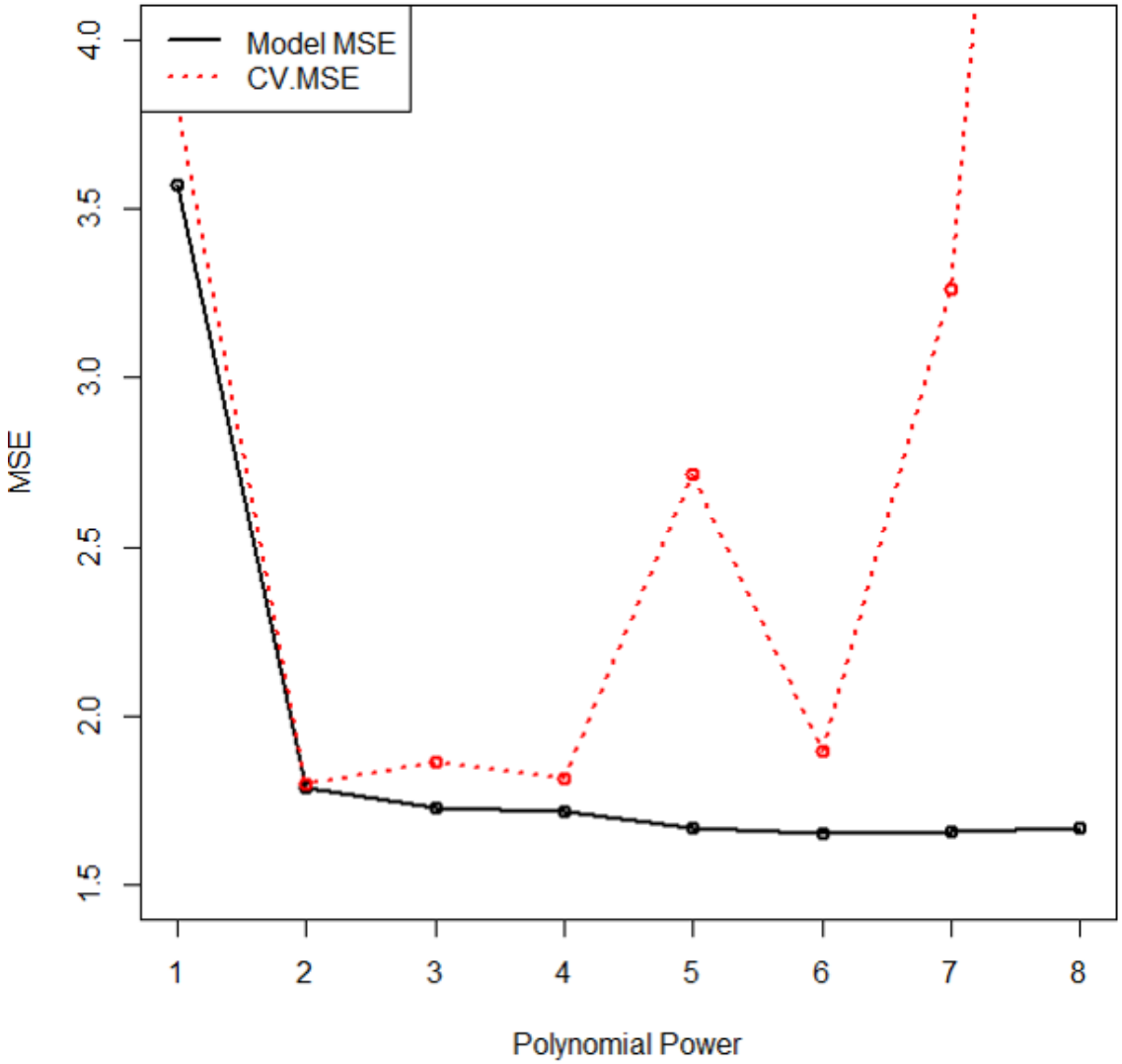
An Example



An Example


Y~poly(X, 1)
Y~poly(X, 2)
Y~poly(X, 3)
Y~poly(X, 4)
Y~poly(X, 5)
Y~poly(X, 6)
Y~poly(X, 7)
Y~poly(X, 8)
0
5
10
-2
-1
0
1
2

An Example


Model MSE
CV.MSE
MSE
4.0
3.5
3.0
2.5
2.0
1.5
1
2
3
4
5
6
7
8
Polynomial Power

Bias-Variance Tradeoff

- This is a very important tradeoff that governs the choice of statistical learning methods.
- **Bias**: how far the estimated model $\hat{f}(x)$ is to the true model $f(x)$.
  - Unbiased estimate is defined as: $\mathbb{E}\hat{f}(x) = f(x)$
  - Usually, we calculate the squared bias: $(\mathbb{E}\hat{f}(x) - f(x))^2$
- **Variance**: the variation of estimated model $\hat{f}(x)$ based on different training set.

Train 1 $\hat{f}_1$

2 $\hat{f}_2$

100

$Var(\hat{f})$ | $(\bar{\hat{f}} - f)$

$(E\hat{f} - f)^2 = Bias^2$

Bias-Variance Tradeoff

Source: link

Bias-Variance Tradeoff

- Suppose the data arise from a model $Y = f(x) + \epsilon$, with $\mathbb{E}(\epsilon) = 0$ and $Var(\epsilon) = \sigma^2$.
- Suppose $\hat{f}(x)$ is trained based on some training data, and let (x_0, y_0) be a test observation from the same population.
- The *expected prediction error* can be decomposed to:

$$\mathbb{E}[y_0 - \hat{f}(x_0)]^2 = \sigma^2 + Bias^2(\hat{f}(x_0)) + Var(\hat{f}(x_0))$$

(Exercise: Show it.)

$y_0 = f(x) + \varepsilon$

$E(\varepsilon + f(x) - \hat{f}(x))^2$

$\sigma^2 + Bias^2 + Var$

- Typically as the **flexibility** of $\hat{f}$ increases, its variance increases, and its bias decreases.

MSPE : Mean Squared <u>Prediction</u> Error

$$E[y_0 - \hat{f}(x_0)]^2 = E[f(x_0) + \varepsilon - \hat{f}(x_0)]^2$$

$$= E(\varepsilon^2) + \underbrace{[E(\hat{f}(x_0)) - f(x_0)]^2}_{= \text{Bias}^2} + \underbrace{E[\hat{f}(x_0) - E\hat{f}(x_0)]^2}_{= \text{Variance}}$$

$$= \sigma^2 + \text{Bias}^2 + \text{Variance}$$

e.g. $(E\hat{\beta} - \beta)^2$

e.g. $E(\hat{\beta} - E\hat{\beta})^2$

$= Var(\varepsilon)$

irreducible Error

Tradeoff

Classification Problems

- Response variable is **qualitative**
- Train a classifier $\hat{C}(x)$ that can label any new input data x
- It usually involves certain decision rule
- Prediction (testing) error: Misclassification rate (MR)

$$MR_{Te} = \frac{1}{M} \sum_{i \in Te} \mathbb{I}[y_i \neq \hat{C}(x_i)]$$

Learning Resources

- Data
  - Textbook data (James, et al.): install R package `ISLR`
  - UCI Machine Learning Repository
  - Kaggle
  - KDD Nuggets
- Lectures and other tutorials
  - Textbook videos (James, et al.)
  - CMU Statistical Machine Learning Course (very theoretical)
  - DataCamp
  - Coursera